The low-level mysteries of pipeline barriers

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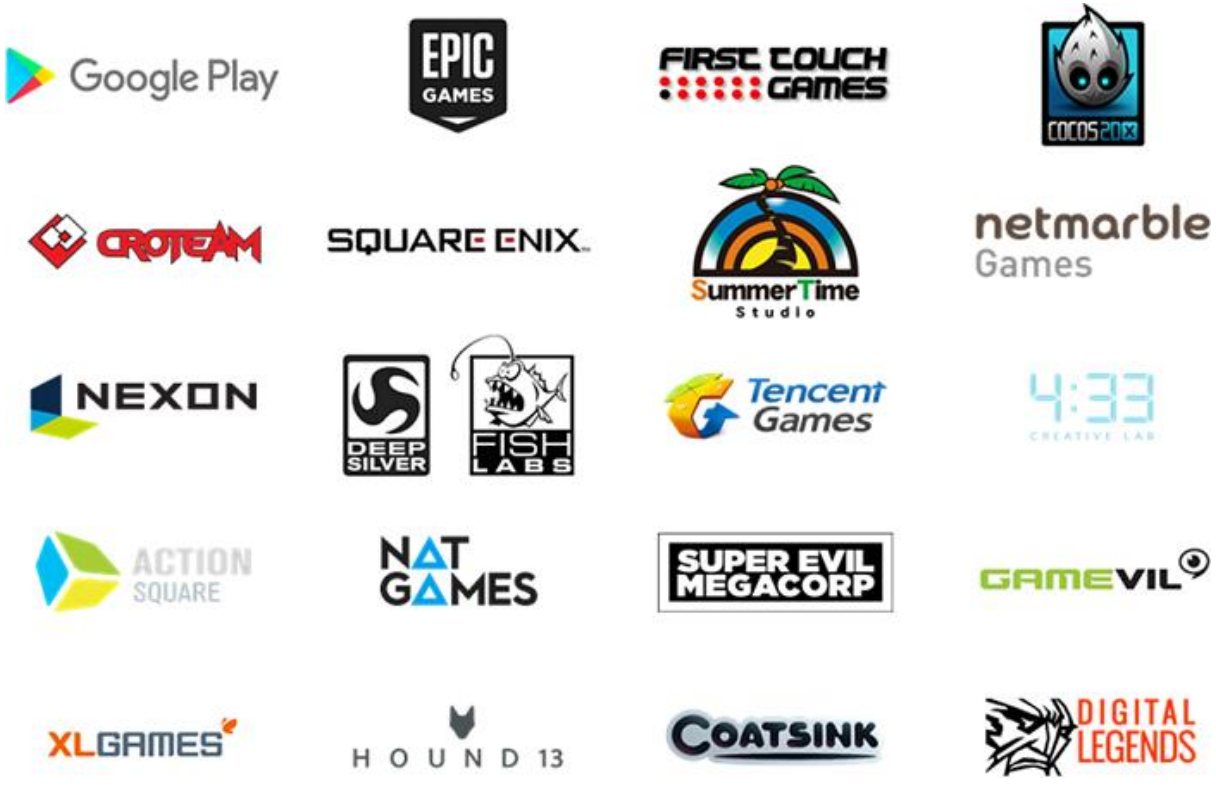
Overview

- Introduction
- What are barriers?
- How are barriers exposed in Vulkan?
- Case study and examples

Our partners



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Who are we?

- Promote the use of Vulkan on Android
- Support game studios with issues on our devices, at a global scale
- Help game studios port their games to Vulkan
  - Performance, content tuning, DDK & platform support

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Why do GPUs need barriers?

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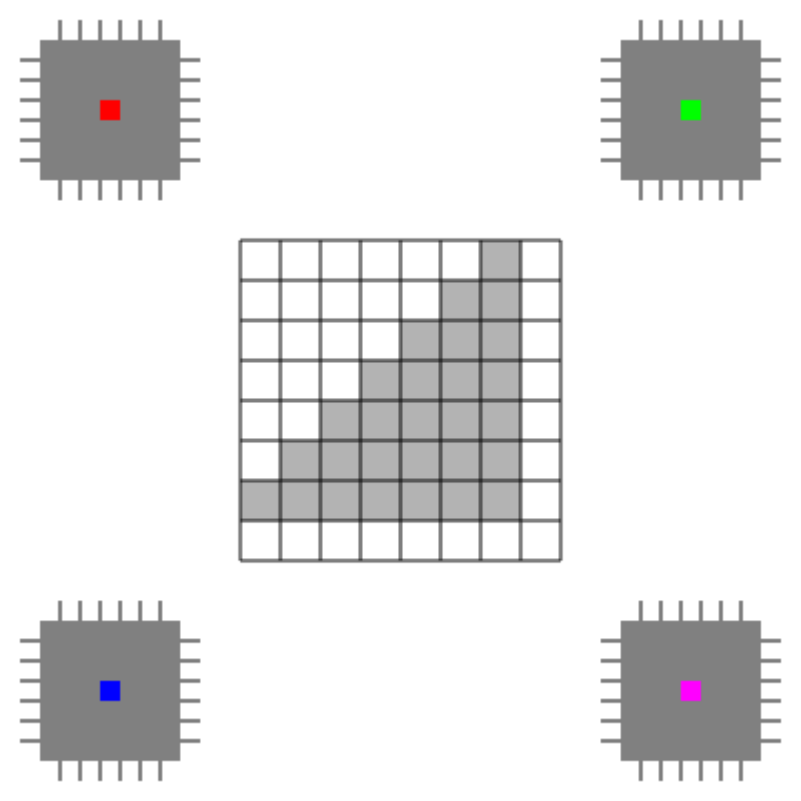
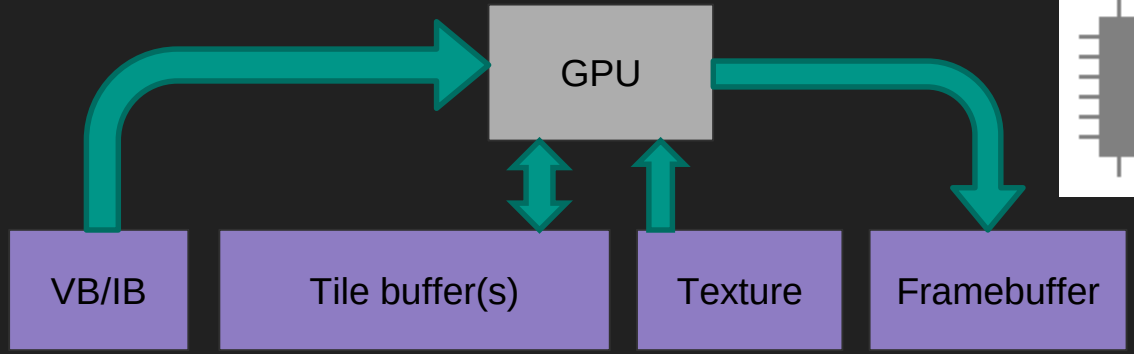
GPUs are highly parallel

- “Computer graphics is embarrassingly parallel”
- “In parallel computing, an **embarrassingly parallel** workload or problem [...] is one where *little or no effort* is needed to separate the problem into a number of parallel tasks.” - Wikipedia
  - Heh, heh, heh
- “All problems become scalar once you’ve thrown enough silicon at them”
  - - A. Garrard, 2018

GPUs are highly parallel

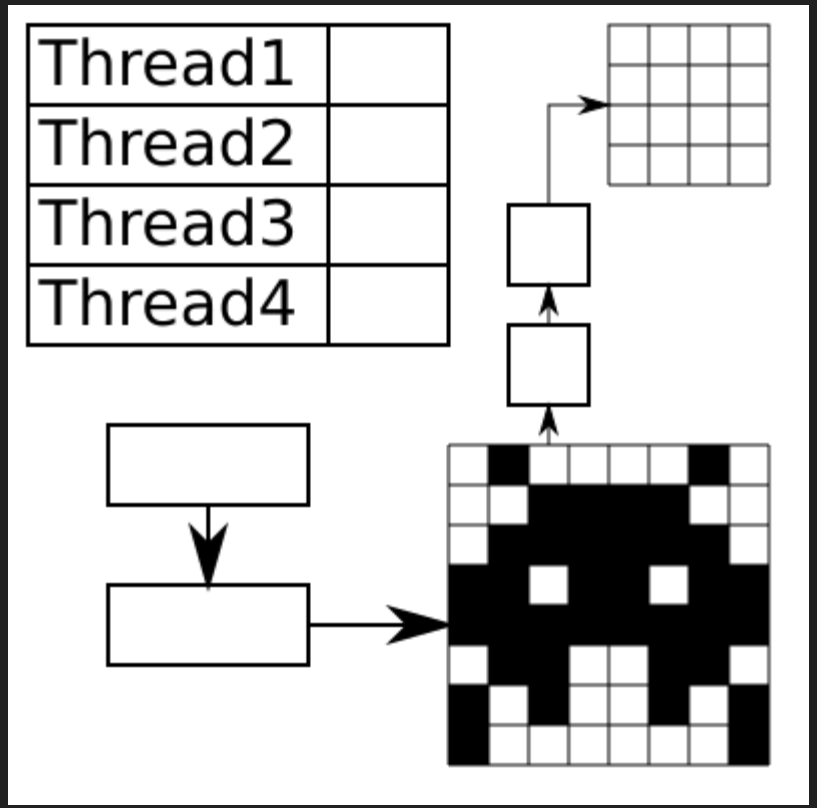
- Rasterisation is quite parallel
- Shading is parallel
- Memory access is parallel
  - Multiple usage-dependent caches and buffers
- All GPUs are parallel
  - But some are more = than others - George Orwe





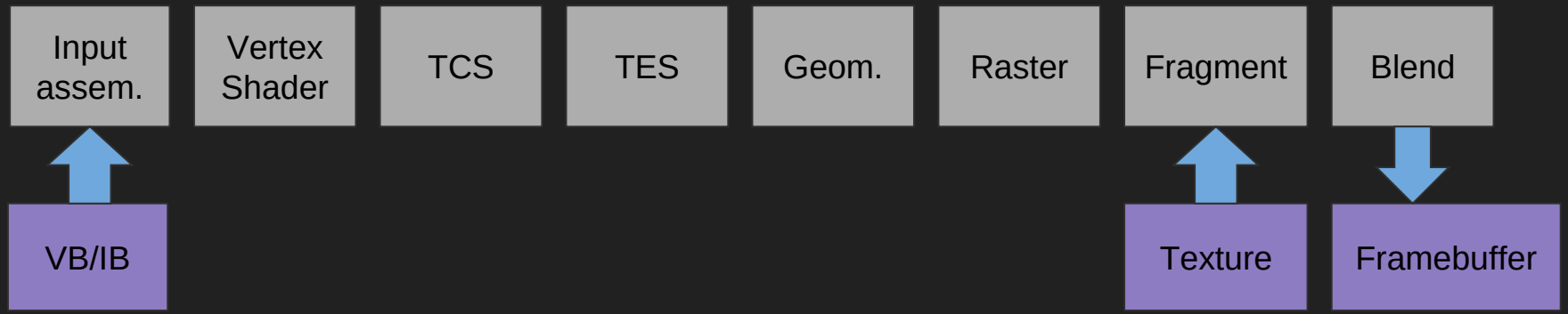
GPUs are hyperthreaded

- Graphics is dominated by memory access
  - Textures, frame buffers, vertex buffers
- Many threads let GPUs hide latency
  - ALUs are often quite deeply pipelined, but memory latency can be enormous
- Even more is in flight than the parallelism would suggest!



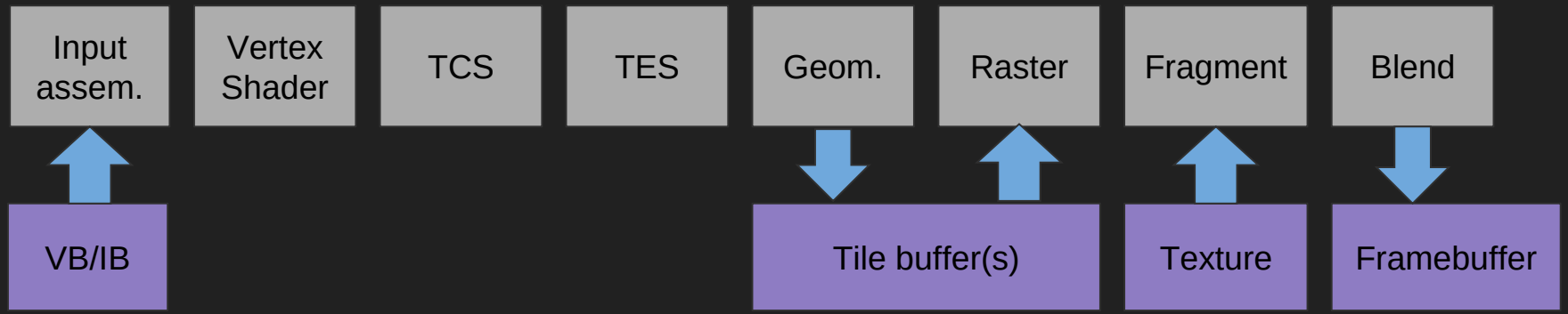
GPUs are heavily pipelined

- The “graphics pipeline” is actually a series of (mostly) parallel stages
- The pipeline can get backed up by slow components, so buffering is important to keep things flowing
- The GPU can be working on multiple elements at once within a stage
- Triangles covering a fragment may not finish shading in-order



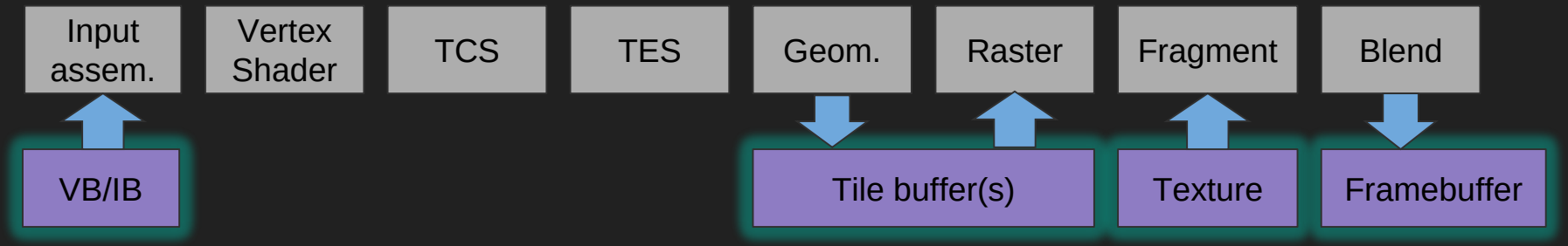
Tiled GPUs are *very* heavily pipelined

- Tiled GPUs do all the rasterising for one tile independently of other tiles
- Fragments in one tile may get shaded before “earlier” primitives in another
- Rasterising may not be in the same frame as vertex shading
- Vertex shaders may be run repeatedly
- “How long did my shader take?” is a very complicated question



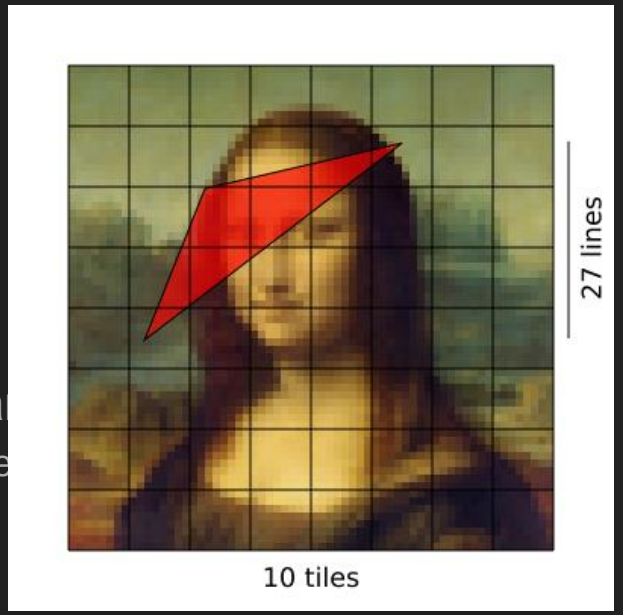
Even the memory is parallel

- GPUs depend heavily on caches
  - Random access to “fast” DRAM is much slower than you’d think
  - 1995: 100MHz SDRAM random access latency: ~20ns
  - 2018: DDR4-4800 random access latency: ~8ns (2.5x in 23 years)
- Textures might be in a special cache
- The frame buffer on a tiler may be a “special” cache
- The caches don’t necessarily snoop each other



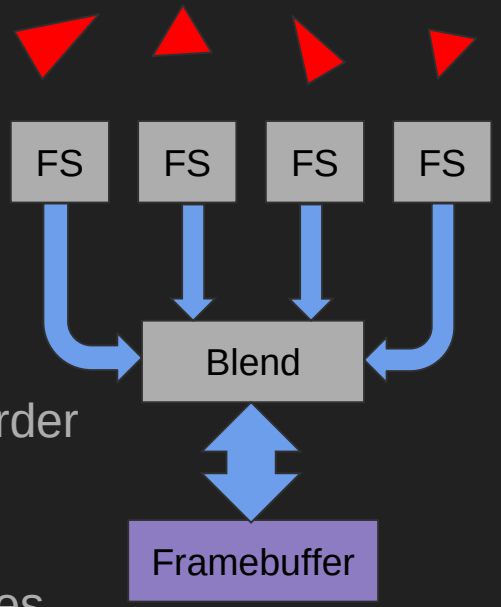
Memory doesn’t look like reality

- Linear memory is really inefficient for texturing
  - Exact layout depends on cache size,
    ease of integration, and patents…
- “Layout” enables memory compression to save ba
  - Lossless compression for the frame buffer, MS, depth buffe
  - Layout is usage-specific
  - The representation may not be consistent (e.g. NaNs)
  - Don’t confuse this with lossy texture compression
    (DXT, ETC, ASTC, PVRTC)
  - Barriers specify layout transitions
- The CPU’s view has to be non-proprietary
  - GPU vendors can’t expose details, because then they wouldn’t be able to change them
  - Still need a simple view for communication with the CPU



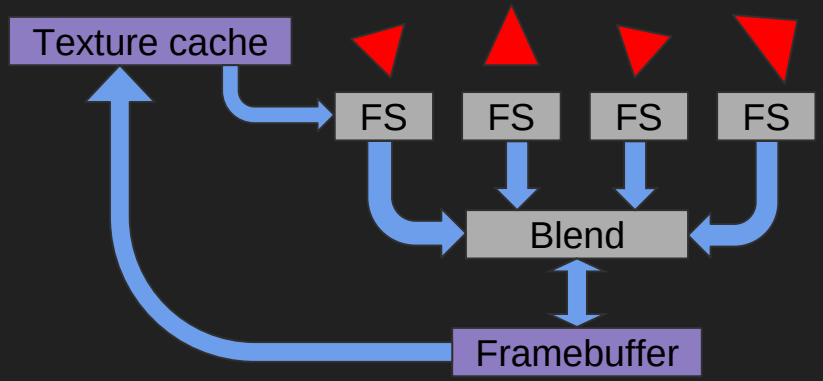
Magic just happens

- If *every* parallel operation needed manual sequencing, game developers would go mad*
  - (* *more* mad)
- The silicon designers get to go mad* instead
  - (* *more* mad)
- GPUs maintain the illusion primitives are rasterised in order
  - Typically blend units can sort out the mess
- ...so long as you're only writing to the frame buffer
- Computer graphics is cheating and hoping no-one notices
  - Corollary: do what you like, but don't get caught
  - - A. Garrard, 2018



There’s no such thing as magic

- Sometimes you do things the GPU can’t magic away
  - Abstracting away parallelism is easier in special cases than in general
  - Older APIs try to apply workarounds heuristically, which can cause unnecessary overhead
- Reading the frame buffer during rendering is hard
  - Requires pixel ordering guarantees, has representation issues
- Writing outside the framebuffer in any kind of shader is not strictly ordered automagically
  - E.g. intermediate outputs from vertex shaders
- Accessing the framebuffer other than the current pixel complicates tiling

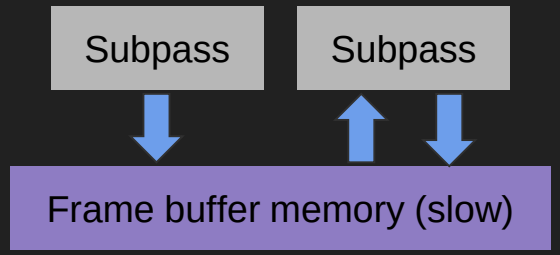


GPGPU/Compute - more than pretty pictures

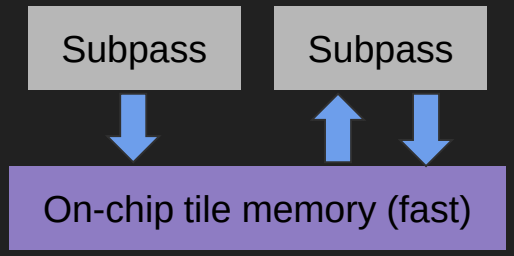
- Programmable GPUs used for more than just graphics since about 2001
- Custom compute shaders and APIs for many years
- More user control over read and write
- Much more requirement on the user to synchronise everything

Local dependencies

- Reads after framebuffer writes can have a *framebuffer local dependency*
- In a tiler, this dependency can stay within the processing of the tile
  - If you want to read anywhere in the framebuffer, you need the whole image to be rendered
  - If you just want to read your current pixel, you can work within the current tile
- Framebuffer local dependencies let you synchronise within the current tile processing rather than across the entire frame
- Local processing can avoid a lot of unnecessary memory traffic



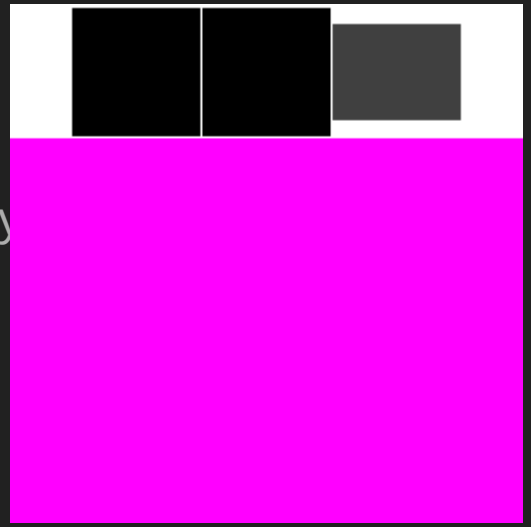
Global access



Local access

Subpass synchronisation - mostly magic

- Typical graphics usage pattern: writing one pixel to the frame buffer, then reading it back in a second subpass
  - Most common for deferred shading and programmable multisample resolves
  - Tilers can stay in tile memory for this
- Local dependencies get handled automatically
- Low synchronisation overhead
- Much less costly than a full framebuffer write to memory
  - FB writes appear much less costly on a desktop GPU
- This is why the subpass concept exists in Vulkan
- Only works for local access
  - Bear this in mind if you’re post-processing



Pipeline barriers

- Block operations after this point until operations before this point complete
- Dependencies apply to graphics pipeline stages
  - Block only the pipeline stages you need (especially on a tiler!)
- Dependencies can be framebuffer local, (but use subpasses!)
- Use between render passes (e.g. shadow map to main frame)
- Use to order compute operations

Events: sync while keeping things busy

- A simple barrier divides time into before and after
  - Work may have to stop and wait, stalling the GPU
  - But we wanted parallelism - how do we keep things going while we synchronise?
  - Doing a single barrier for multiple dependencies helps - but still stalls
- Events let you have multiple dependencies active
  - Wait only for the work you cared about, independent work can continue
- Events have a user-visible representation
  - The host CPU can access them too
  - Be careful not to make the GPU time out when doing this

Summary

- GPUs do lots of things at once
  - They get a lot slower if they can’t do this!
- Only the most basic ordering happens automatically
- Anything more complicated, you need to provide explicit synchronisation
- Use:
  - Subpass dependencies for local pixel framebuffer dependencies
  - Pipeline barriers to synchronise everything else
  - Events to keep the GPU busy while synchronising
- What’s actually going on may well be more complicated than this
- That’s the theory - how do you program it?

Frederic Garnier, Samsung Electronics

How are barriers exposed in Vulkan?

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Disclaimer

- Results and our experience are based on Galaxy S7 to S9 devices using Arm Mali and Qualcomm GPUs

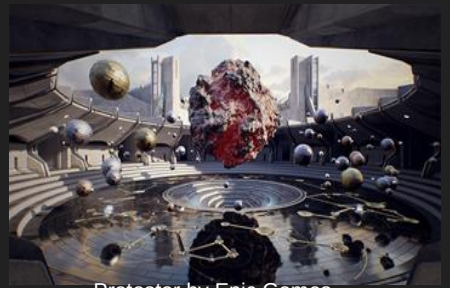
AxE by Nexon

FF15 by Square Enix

L2R by Netmarble

Arena of Valor by Tencent

Blade II by Action Square


Protostar by Epic Games

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How are barriers exposed in Vulkan?

- Need to track resources in Vulkan and synchronise accordingly
  - Is the resource in the right state?
  - Are we writing to / reading to the resource in the correct order?
  - Have we taken care of execution and memory dependencies?
- Even lower-level…
  - Have we ensured that data is visible and available to the relevant stages?

How are barriers exposed in Vulkan?

```
void vkCmdPipelineBarrier(
    VkCommandBuffer          commandBuffer,
    VkPipelineStageFlags     srcStageMask,
    VkPipelineStageFlags     dstStageMask,
    VkDependencyFlags        dependencyFlags,
    uint32_t                 memoryBarrierCount,
    const VkMemoryBarrier*   pMemoryBarriers,
    uint32_t                 bufferMemoryBarrierCount,
    const VkBufferMemoryBarrier*  pBufferMemoryBarriers,
    uint32_t                     imageMemoryBarrierCount,
    const VkImageMemoryBarrier*   pImageMemoryBarriers);
```

```
typedef enum VkPipelineStageFlagBits {
    VK_PIPELINE_STAGE_TOP_OF_PIPE_BIT = 0x00000001,
    VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT = 0x00000002,
    VK_PIPELINE_STAGE_VERTEX_INPUT_BIT = 0x00000004,
    VK_PIPELINE_STAGE_VERTEX_SHADER_BIT = 0x00000008,
    VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT = 0x00000010,
    VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT = 0x00000020,
    VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT = 0x00000040,
    VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT = 0x00000080,
    VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT = 0x00000100,
    VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT = 0x00000200,
    VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT = 0x00000400,
    VK_PIPELINE_STAGE_COMPUTE_SHADER_BIT = 0x00000800,
    VK_PIPELINE_STAGE_TRANSFER_BIT = 0x00001000,
    VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT = 0x00002000,
    VK_PIPELINE_STAGE_HOST_BIT = 0x00004000,
    VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT = 0x00008000,
    VK_PIPELINE_STAGE_ALL_COMMANDS_BIT = 0x00010000
} VkPipelineStageFlagBits;
```

How are barriers exposed in Vulkan?

```
void vkCmdPipelineBarrier(
    VkCommandBuffer          commandBuffer,
    VkPipelineStageFlags     srcStageMask,
    VkPipelineStageFlags     dstStageMask,
    VkDependencyFlags        dependencyFlags,
    uint32_t                 memoryBarrierCount,
    const VkMemoryBarrier*   pMemoryBarriers,
    uint32_t                 bufferMemoryBarrierCount,
    const VkBufferMemoryBarrier*  pBufferMemoryBarriers,
    uint32_t                     imageMemoryBarrierCount,
    const VkImageMemoryBarrier*   pImageMemoryBarriers);
```

```
typedef enum VkDependencyFlagBits {
    VK_DEPENDENCY_BY_REGION_BIT = 0x00000001;
} VkDependencyFlagBits;
```

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How are barriers exposed in Vulkan?

```
void vkCmdPipelineBarrier(
    VkCommandBuffer          commandBuffer,
    VkPipelineStageFlags     srcStageMask,
    VkPipelineStageFlags     dstStageMask,
    VkDependencyFlags        dependencyFlags,
    uint32_t                 memoryBarrierCount,
    const VkMemoryBarrier*   pMemoryBarriers,
    uint32_t                 bufferMemoryBarrierCount,
    const VkBufferMemoryBarrier*  pBufferMemoryBarriers,
    uint32_t                     imageMemoryBarrierCount,
    const VkImageMemoryBarrier*   pImageMemoryBarriers);
```

```
typedef struct VkImageMemoryBarrier {
    VkStructureType            sType;
    const void*                pNext;
    VkAccessFlags              srcAccessMask;
    VkAccessFlags              dstAccessMask;
    VkImageLayout              oldLayout;
    VkImageLayout              newLayout;
    uint32_t                   srcQueueFamilyIndex;
    uint32_t                   dstQueueFamilyIndex;
    VkImage                    image;
    VkImageSubresourceRange    subresourceRange;
} VkImageMemoryBarrier;
```

How are barriers exposed in Vulkan?

```
typedef struct VkImageMemoryBarrier {
    VkStructureType            sType;
    const void*                pNext;
    VkAccessFlags              srcAccessMask;
    VkAccessFlags              dstAccessMask;
    VkImageLayout              oldLayout;
    VkImageLayout              newLayout;
    uint32_t                   srcQueueFamilyIndex;
    uint32_t                   dstQueueFamilyIndex;
    VkImage                    image;
    VkImageSubresourceRange    subresourceRange;
} VkImageMemoryBarrier;
```

```
typedef enum VkAccessFlagBits {
    VK_ACCESS_INDIRECT_COMMAND_READ_BIT = 0x00000001,
    VK_ACCESS_INDEX_READ_BIT = 0x00000002,
    VK_ACCESS_VERTEX_ATTRIBUTE_READ_BIT = 0x00000004,
    VK_ACCESS_UNIFORM_READ_BIT = 0x00000008,
    VK_ACCESS_INPUT_ATTACHMENT_READ_BIT = 0x00000010,
    VK_ACCESS_SHADER_READ_BIT = 0x00000020,
    VK_ACCESS_SHADER_WRITE_BIT = 0x00000040,
    VK_ACCESS_COLOR_ATTACHMENT_READ_BIT = 0x00000080,
    VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT = 0x00000100,
    VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT = 0x00000200,
    VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT = 0x00000400,
    VK_ACCESS_TRANSFER_READ_BIT = 0x00000800,
    VK_ACCESS_TRANSFER_WRITE_BIT = 0x00001000,
    VK_ACCESS_HOST_READ_BIT = 0x00002000,
    VK_ACCESS_HOST_WRITE_BIT = 0x00004000,
    VK_ACCESS_MEMORY_READ_BIT = 0x00008000,
    VK_ACCESS_MEMORY_WRITE_BIT = 0x00010000,
} VkAccessFlagBits;
```

How are barriers exposed in Vulkan?

```
typedef struct VkImageMemoryBarrier {
    VkStructureType            sType;
    const void*                pNext;
    VkAccessFlags              srcAccessMask;
    VkAccessFlags              dstAccessMask;
    VkImageLayout              oldLayout;
    VkImageLayout              newLayout;
    uint32_t                   srcQueueFamilyIndex;
    uint32_t                   dstQueueFamilyIndex;
    VkImage                    image;
    VkImageSubresourceRange    subresourceRange;
} VkImageMemoryBarrier;
```

```
typedef enum VkImageLayout {
    VK_IMAGE_LAYOUT_UNDEFINED = 0,
    VK_IMAGE_LAYOUT_GENERAL = 1,
    VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL = 2,
    VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL = 3,
    VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL = 4,
    VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL = 5,
    VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL = 6,
    VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL = 7,
    VK_IMAGE_LAYOUT_PREINITIALIZED = 8,
    VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL = 1000117000,
    VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL = 1000117001,
    VK_IMAGE_LAYOUT_PRESENT_SRC_KHR = 1000001002,
} VkImageLayout;
```

How are barriers exposed in Vulkan?

- Images in Vulkan are created with a tiling arrangement
  - Linear tiling
  - Optimal aka swizzled tiling
- Images that are created with optimal tiling require an explicit copy op
  - Possible to avoid this copy if using linear tiling mode
  - Useful if the texture is streamed in every frame...
- But images with linear tiling have a lot of limitations
  - No support for mipmaps
  - Only a few formats may be supported…

How are barriers exposed in Vulkan?

- Will go through pipeline barriers using the following example
  - Copy data to an optimal image from a buffer or a linear image that contains data…
  - Synchronize correctly to prepare the implementation for the copy operation...
  - Synchronize correctly to prepare the implementation for sampling the copied image…

Copying data to an image

```
void vkCmdPipelineBarrier(
    commandBuffer,
    VK_PIPELINE_STAGE_TOP_OF_PIPE_BIT,
    VK_PIPELINE_STAGE_TRANSFER_BIT,
    0,
    0
    VK_NULL_HANDLE,
    0,
    VK_NULL_HANDLE,
    1,
    &imageBarrier1;
```

```
imageBarrier1 = {
    VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER,
    VK_NULL_HANDLE,
    0,
    VK_ACCESS_TRANSFER_WRITE_BIT,
    VK_IMAGE_LAYOUT_UNDEFINED,
    VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL,
    VK_QUEUE_FAMILY_IGNORED,
    VK_QUEUE_FAMILY_IGNORED,
    imageHandle,
    subResourcesRange;
};
```

Sampling data from a copied-to image

```
void vkCmdPipelineBarrier(
    commandBuffer,
    VK_PIPELINE_STAGE_TRANSFER_BIT,
    VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT,
    0,
    0
    VK_NULL_HANDLE,
    0,
    VK_NULL_HANDLE,
    1,
    &imageBarrier2;
```

```
imageBarrier2 = {
    VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER,
    VK_NULL_HANDLE,
    VK_ACCESS_TRANSFER_WRITE_BIT,
    VK_ACCESS_SHADER_READ_BIT,
    VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL,
    VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL,
    VK_QUEUE_FAMILY_IGNORED,
    VK_QUEUE_FAMILY_IGNORED,
    imageHandle,
    subResourcesRange;
};
```

Batch your pipeline barriers!

- Inserting a pipeline barrier (sync-point) within the command buffer has a CPU cost associated to it
  - Need to batch barriers as much as possible and flush at the right time!

```
vkCmdPipelineBarrier(...,1,&barrier)
vkCmdPipelineBarrier(...,1,&barrier)
vkCmdPipelineBarrier(...,1,&barrier)
vkCmdPipelineBarrier(...,1,&barrier)
vkCmdPipelineBarrier(...,1,&barrier)
vkCmdPipelineBarrier(...,1,&barrier)
vkCmdPipelineBarrier(...,1,&barrier)
vkCmdPipelineBarrier(...,1,&barrier)
vkCmdPipelineBarrier(...,1,&barrier)
```

```
vkCmdPipelineBarrier(...,9, barriers.getData())
```

Subpass dependencies

- Images also need to be transitioned to the correct layout before presentation
- Preferably to transition as part of the render pass if possible
  - Can specify an image layout to use per-subpass and a final layout
  - Final layout is what the image transitions to at the end of the render pass
- Why not use a subpass dependency for the previous case?
  - Due to render pass scope .. copy command can only be called outside of a render pass instance

Subpass dependencies

```
typedef struct VkAttachmentReference {
    uint32_t        attachment;
    VkImageLayout   layout;
} VkAttachmentReference;

typedef struct VkAttachmentDescription {
    VkAttachmentDescriptionFlags    flags;
    VkFormat                        format;
    VkSampleCountFlagBits           samples;
    VkAttachmentLoadOp              loadOp;
    VkAttachmentStoreOp             storeOp;
    VkAttachmentLoadOp              stencilLoadOp;
    VkAttachmentStoreOp             stencilStoreOp;
    VkImageLayout                   initialLayout;
    VkImageLayout                   finalLayout;
} VkAttachmentDescription;
```

```
typedef struct VkSubpassDependency {
    uint32_t                srcSubpass;
    uint32_t                dstSubpass;
    VkPipelineStageFlags    srcStageMask;
    VkPipelineStageFlags    dstStageMask;
    VkAccessFlags           srcAccessMask;
    VkAccessFlags           dstAccessMask;
    VkDependencyFlags       dependencyFlags;
} VkSubpassDependency;
```

Subpass dependencies

- Very simple example based on rendering a triangle or quad and presenting it
- Transition at the beginning of a render pass instance may happen out of order
  - Need to make sure presentation engine is done reading from the image
  - Subpass dependencies allow us to express execution and memory dependencies we need
  - Implicit subpass dependencies exist but not suitable for this use case

Subpass dependencies

- Transition the image when it can be rendered to..
  - I.e. when made available by semaphore & based on pWaitDstStageMask
- Not just limited to synchronising with presentation engine
  - Render passes can be used for off-screen rendering
  - Next one depends on previous one
  - Stage masks and access flags need to be set accordingly

Subpass dependencies - Implicit pre-dependency

```
colorAttachmentReference = {
    0,
    VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
};

attachmentDescription = {
    0,
    VK_FORMAT_R8G8B8_UNORM,
    VK_SAMPLE_COUNT_1_BIT,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE,
    VK_ATTACHMENT_STORE_OP_STORE,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE,
    VK_ATTACHMENT_STORE_OP_DONT_CARE,
    VK_IMAGE_LAYOUT_UNDEFINED,
    VK_IMAGE_LAYOUT_PRESENT_SRC_KHR
};
```

```
subpassDependency = {
    VK_SUBPASS_EXTERNAL,
    firstSubpass,
    VK_PIPELINE_STAGE_TOP_OF_PIPE_BIT,
    VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
    0,
VK_ACCESS_INPUT_ATTACHMENT_READ_BIT |
VK_ACCESS_COLOR_ATTACHMENT_READ_BIT |
VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT |
VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT
|
VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT
,
    0
};
```

Subpass dependencies - Explicit dependency

```
colorAttachmentReference = {
    0,
    VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
};

attachmentDescription = {
    0,
    VK_FORMAT_R8G8B8_UNORM,
    VK_SAMPLE_COUNT_1_BIT,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE,
    VK_ATTACHMENT_STORE_OP_STORE,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE,
    VK_ATTACHMENT_STORE_OP_DONT_CARE,
    VK_IMAGE_LAYOUT_UNDEFINED,
    VK_IMAGE_LAYOUT_PRESENT_SRC_KHR
};
```

```
subpassDependency = {
    VK_SUBPASS_EXTERNAL,
    0,
    VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT,
    VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT,
    0,
VK_ACCESS_COLOR_ATTACHMENT_READ_BIT |
VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT |
    0
};
```

Subpass dependencies

- Need a final layout transition to prepare for presentation…
  - Images need to be in *VK_IMAGE_LAYOUT_PRESENT_SRC_KHR* layout
- An implicit post-render pass dependency also exists
  - We don’t need to explicitly define this
  - Defines that the transition happens after all work is done aka bottom of pipe
  - Semaphore guarantees execution dependency for us...

Subpass dependencies - Implicit post-dependency

```
colorAttachmentReference = {
    0,
    VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
};

attachmentDescription = {
    0,
    VK_FORMAT_R8G8B8_UNORM,
    VK_SAMPLE_COUNT_1_BIT,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE,
    VK_ATTACHMENT_STORE_OP_STORE,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE,
    VK_ATTACHMENT_STORE_OP_DONT_CARE,
    VK_IMAGE_LAYOUT_UNDEFINED,
    VK_IMAGE_LAYOUT_PRESENT_SRC_KHR
};
```

```
subpassDependency = {
    lastSubpass,
    VK_SUBPASS_EXTERNAL,
    VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
    VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT,
VK_ACCESS_INPUT_ATTACHMENT_READ_BIT |
VK_ACCESS_COLOR_ATTACHMENT_READ_BIT |
VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT |
VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT
|
VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT
,
    0,
    0
};
```

Subpass dependencies - Explicit dependency

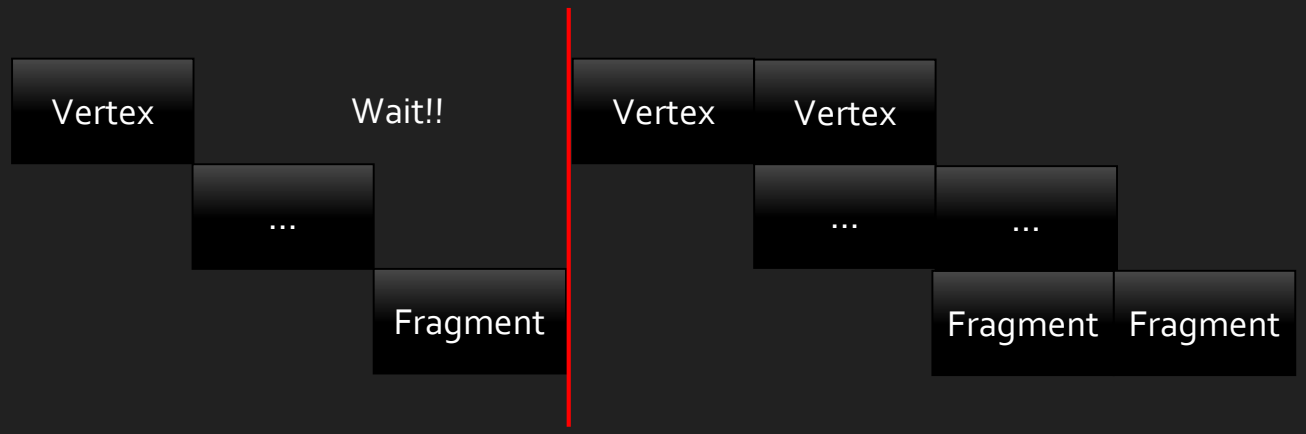
```
colorAttachmentReference = {
    0,
    VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
};

attachmentDescription = {
    0,
    VK_FORMAT_R8G8B8_UNORM,
    VK_SAMPLE_COUNT_1_BIT,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE,
    VK_ATTACHMENT_STORE_OP_STORE,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE,
    VK_ATTACHMENT_STORE_OP_DONT_CARE,
    VK_IMAGE_LAYOUT_UNDEFINED,
    VK_IMAGE_LAYOUT_PRESENT_SRC_KHR
};
```

```
subpassDependency = {
    VK_SUBPASS_EXTERNAL,
    0,
    VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT,
    VK_PIPELINE_BOTTOM_OF_PIPE_BIT,
VK_ACCESS_COLOR_ATTACHMENT_READ_BIT |
VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT |
    0,
    0
};
```

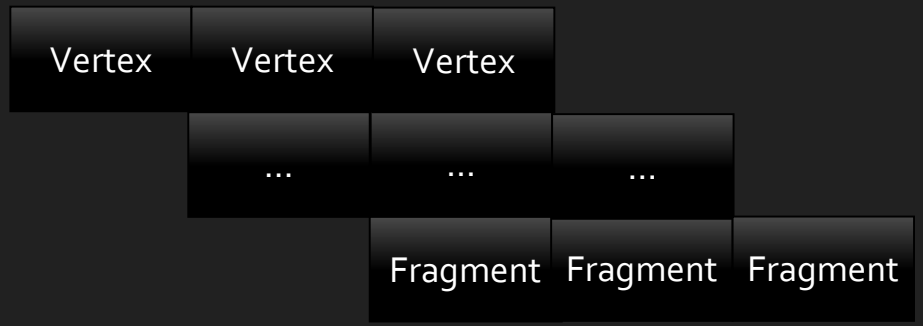
Case Study / Examples

- Transitioning image to a readable state… but using wrong stages
  - *srcStageMask = FRAGMENT_SHADER_BIT*
  - *dstStageMask = VERTEX_SHADER_BIT | VERTEX_INPUT_BIT*



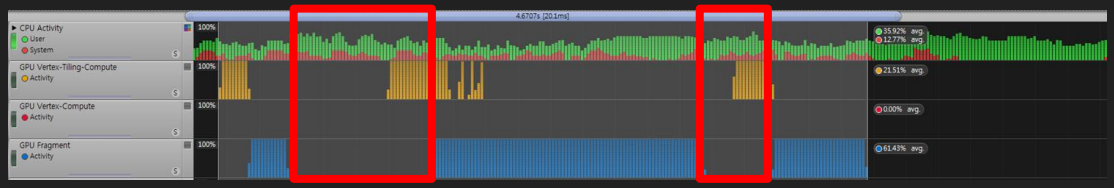
Case Study / Examples

- Transitioning image to a readable state… but using wrong stages
  - *srcStageMask = COLOR_ATTACHMENT_OUTPUT_BIT*
  - *dstStageMask = FRAGMENT_SHADER_BIT*

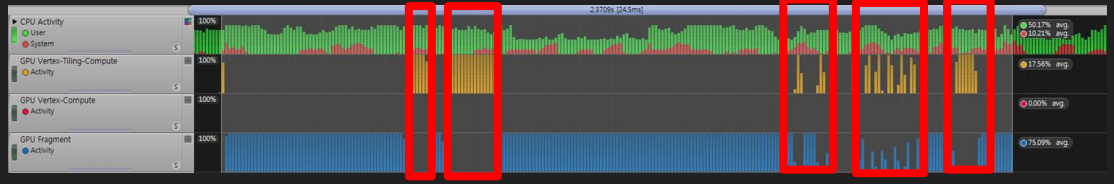


Case Study / Examples

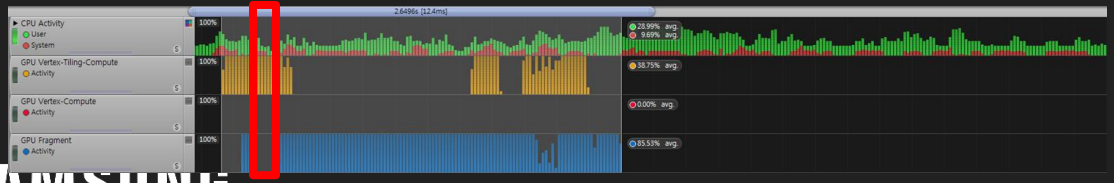
L2R GLES build - 20.1ms frame time



L2R Vulkan (incorrect barrier stages) - 24.5ms (+3.4)



L2R Vulkan (incorrect barrier stages) - 12.4ms (-8.3)



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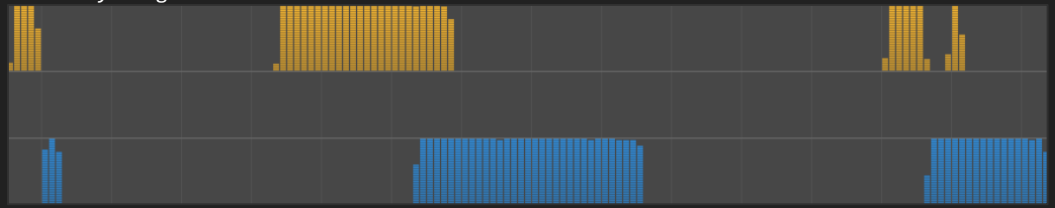
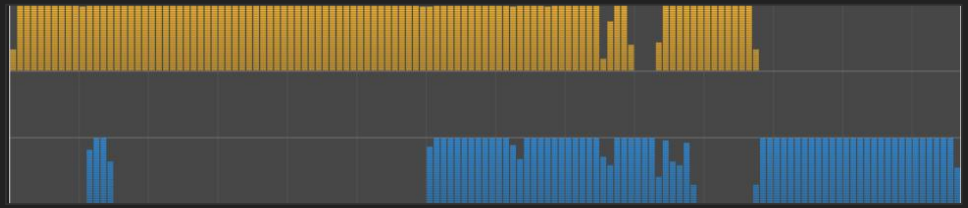
Case Study / Examples

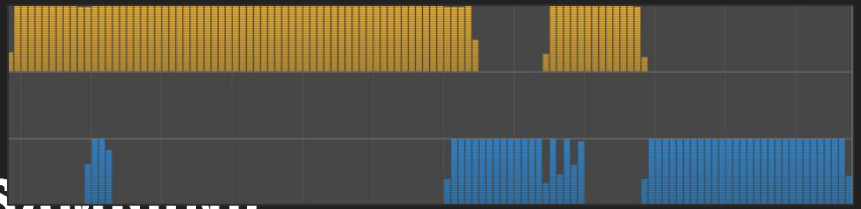
Unity Original GLES - 14.9ms



Unity VK - Using ALL_GRAPHICS_BIT - 13.7ms (-1.2)

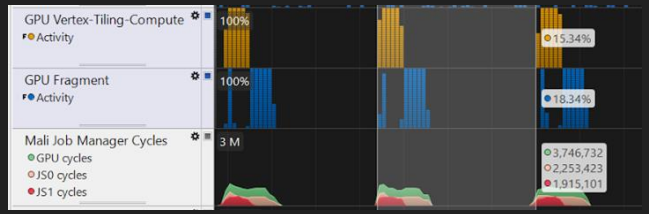


Unity VK - Using optimized barriers + batching (-2.9)

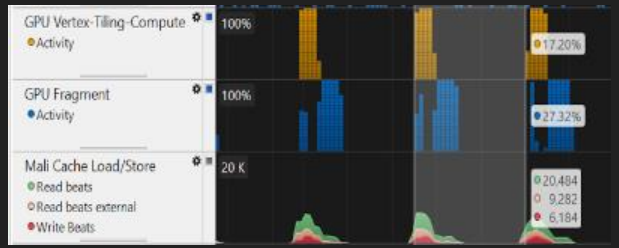


Case Study / Examples

Original build - 36ms frame time



Optimized build - 25ms (-11) frame time



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